

Lab Report – Goal-Based Warehouse Agent with an LLM

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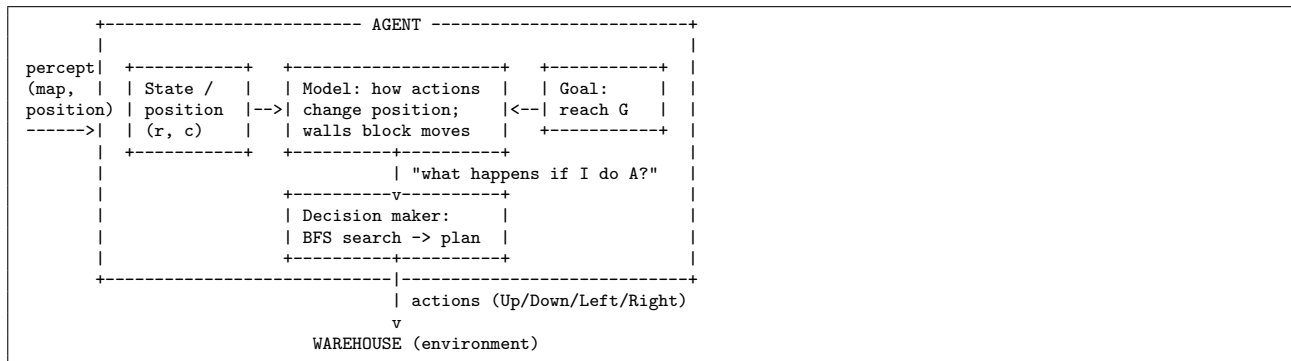
Task 1: Understanding the problem

1. **Environment:** a 7×21 grid warehouse; # cells are impassable shelving, . cells are free; fully observable, static, deterministic, discrete, single agent.
2. **Goal:** move the vehicle from S (row 1, col 1) to G (row 1, col 19) along a collision-free path.
3. **Actions:** Up, Down, Left, Right; each moves one cell and is legal only if the target cell is inside the grid and not #.
4. **Information maintained:** the map (model of the world), the current position (state), the goal position, and the search bookkeeping (frontier, visited set, parent links) needed to build the plan.
5. **Goal-based, not simple reflex:** a reflex agent maps the current percept to an action by fixed rules (e.g. “if free on the right, go right”) and would get stuck in dead ends; this agent holds an explicit goal and considers the future consequences of action sequences (search) before acting.

Think about it (twice as large): BFS is still correct and optimal, but its frontier grows with the number of cells (memory $O(|cells|)$ and time roughly $4\times$ for a doubled area), so larger or weighted maps call for A* with a heuristic (Manhattan distance), which expands far fewer nodes. Further difficulties: moving obstacles or other vehicles, partial observability, non-uniform move costs, and replanning when the map changes.

Task 2: Agent design

- **Environment:** warehouse grid. **State:** current cell (r, c) . **Goal:** cell G. **Actions:** Up/Down/Left/Right. **Decision-making component:** BFS planner that searches the transition model for an action sequence reaching the goal.



Task 3: Prompt engineering

Prompt used (the lab’s suggested prompt, plus the exact map and testing requirements):

```
Write a well-documented Python program implementing a goal-based agent for the
warehouse navigation problem shown below.
Map (S start, G goal, # obstacle, . free; moves Up/Down/Left/Right, one cell each):
#####
#S...#.....G#
#.#...#####..#
#...#.....#
#.....#.....#
#.....#.....#
#####
The program should
- represent the warehouse as a two-dimensional grid;
```

```
- determine a collision-free path from S to G;  
- avoid all obstacles;  
- print either the path found or a suitable message if no path exists;  
- explain the search algorithm that has been chosen and why it is appropriate.  
Also write a small test file that checks the path is valid, that its length is  
shortest, and that an unreachable goal prints the no-path message.
```

The code was produced with LLM assistance from this prompt.

Test results. `warehouse_agent.py` ran without errors; it found a 20-move path (Right x3, Down, Right x3, Up, Right x12) which avoids all #. `test_agent.py` confirmed: path ends on G, never enters an obstacle, every step is one cell; its length equals an independently computed shortest distance (20); and a blocked map prints “No collision-free path exists”. Path drawn on the map:

```
#####  
#S***.#####G#  
#.#####..  
#...##.....#  
#.#####.###.#.  
#.....#.....#  
#####
```

Questions.

1. **Working on first attempt?** Yes: the first generated version ran and passed all tests, so no fix iteration was needed. It was not edited; it was verified with the tests above rather than by trusting the printed path.
2. **Improving the prompt anyway:** it helped to give the exact map, name the required tests (validity, optimality, no-path case), and, if needed, specify the output format, the language version, and “do not use external libraries”. A vaguer prompt (“write a program”) leaves the algorithm, map format and output undefined.
3. **Algorithm:** Breadth-First Search over grid cells, with a visited/parent dictionary and path reconstruction.
4. **Why:** all moves cost 1 and the state space is tiny, so BFS is complete and returns a shortest path; it is simple and needs no heuristic. (A* would also work and scales better; Dijkstra would matter only with varying costs.) The prompt’s words “determine a collision-free path” on an unweighted grid point to BFS.

Strengths and limitations of LLM-assisted development

Strength: fast, well-structured, documented code from a precise specification. Limitation: the printed answer looks plausible whether or not it is correct, so independent tests (obstacle check, optimality check, unreachable case) are what establish correctness; the LLM also picks the algorithm implicitly, so the human must check the choice suits the problem (and that it would scale).

Code

`warehouse_agent.py`

```
"""Goal-based agent for the warehouse navigation problem.  
  
The agent keeps a model of the warehouse (a 2-D grid), an explicit goal (cell G)  
and a set of actions (Up, Down, Left, Right). To decide what to do it searches  
ahead with Breadth-First Search (BFS) for a sequence of actions that reaches the  
goal, then executes that plan.  
  
Why BFS: every move costs 1, so BFS is complete (finds a path if one exists) and  
optimal (returns a shortest path). Time/space are O(rows*cols) for this grid.  
"""  
from collections import deque  
  
WAREHOUSE = [  
    "#####",  
    "#S...#.....G#",  
    "#.#####..  
    "#...##.....#",  
    "#.#####.###.#",  
    "#.....#.....#",  
    "#####",  
]  
  
# action name -> (row change, column change)  
ACTIONS = {"Up": (-1, 0), "Down": (1, 0), "Left": (0, -1), "Right": (0, 1)}
```

```

def parse(layout):
    """Return (grid, start, goal) from a list of strings."""
    grid = [list(row) for row in layout]
    start = goal = None
    for r, row in enumerate(grid):
        for c, ch in enumerate(row):
            if ch == "S":
                start = (r, c)
            elif ch == "G":
                goal = (r, c)
    if start is None or goal is None:
        raise ValueError("map must contain both S and G")
    return grid, start, goal

def successors(grid, state):
    """Legal (action, next_state) pairs: stay on the grid and avoid '#'."""
    r, c = state
    for name, (dr, dc) in ACTIONS.items():
        nr, nc = r + dr, c + dc
        if 0 <= nr < len(grid) and 0 <= nc < len(grid[0]) and grid[nr][nc] != "#":
            yield name, (nr, nc)

def bfs(grid, start, goal):
    """Return the list of actions from start to goal, or None if unreachable."""
    frontier = deque([start])
    came_from = {start: None} # state -> (previous state, action); also the visited set
    while frontier:
        state = frontier.popleft()
        if state == goal: # goal test
            plan = []
            while came_from[state] is not None:
                state, action = came_from[state]
                plan.append(action)
            return plan[::-1]
        for action, nxt in successors(grid, state):
            if nxt not in came_from:
                came_from[nxt] = (state, action)
                frontier.append(nxt)
    return None

def follow(start, plan):
    """Execute a plan and return the visited cells (including start)."""
    cells = [start]
    r, c = start
    for a in plan:
        dr, dc = ACTIONS[a]
        r, c = r + dr, c + dc
        cells.append((r, c))
    return cells

def render(grid, cells):
    """Draw the path with '*' on top of the map."""
    out = [row[:] for row in grid]
    for r, c in cells[1:-1]:
        out[r][c] = "*"
    return "\n".join("".join(row) for row in out)

def run(layout):
    grid, start, goal = parse(layout)
    plan = bfs(grid, start, goal)
    if plan is None:
        print("No collision-free path exists from S to G.")
        return None
    cells = follow(start, plan)
    print(f"Path found: {len(plan)} moves")
    print("Actions:", " ".join(plan))
    print("Cells:", cells)
    print(render(grid, cells))
    return plan

if __name__ == "__main__":
    run(WAREHOUSE)

```

test_agent.py

```

from warehouse_agent import *

grid, s, g = parse(WAREHOUSE)
plan = bfs(grid, s, g)
cells = follow(s, plan)
assert cells[-1] == g
assert all(grid[r][c] != "#" for r, c in cells), "path crosses an obstacle"

```

```

assert all(abs(a[0]-b[0]) + abs(a[1]-b[1]) == 1 for a, b in zip(cells, cells[1:]))
print("path valid, length", len(plan))

# independent check of optimality: relax distances with Bellman-Ford style sweep
INF = 10**9
d = {(r, c): INF for r in range(len(grid)) for c in range(len(grid[0])) if grid[r][c] != "#"}
d[s] = 0
changed = True
while changed:
    changed = False
    for (r, c), v in list(d.items()):
        for _, n in successors(grid, (r, c)):
            if v + 1 < d[n]:
                d[n] = v + 1
                changed = True
assert d[g] == len(plan)
print("BFS length equals independent shortest distance:", d[g])

# unreachable goal -> message, None
blocked = ["#####", "#S#G#", "#####"]
assert run(blocked) is None
print("no-path case handled")

```